

Vidit Gupta

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EDUCATION

Dwarkadas J. Sanghvi College of Engineering

Bachelor of Technology in Information Technology

GPA: 9.1

Sep. 2024 – 2028

Jawahar Navodaya Vidyalaya

Secondary and High School

10th: 95.4% 12th: 92%

Jul. 2016 – Apr 2024

RESEARCH EXPERIENCE

Collaborative Researcher

Aug 2026 – Present

Advanced Vision Labs

Remote

- Collaborating on research in representation learning, investigating the geometry and structure of learned representations in modern deep neural networks.
- Reproducing and experimentally analyzing recent representation-learning methods, studying the effects of normalization and feature alignment across pretrained models.

Undergraduate Research Assistant

May 2026 – Present

IIT Roorkee

Remote

- Conducting research on geospatial time-series forecasting for district-level drought prediction across India using a proprietary multi-year dataset.
- Surveyed recent literature on spatiotemporal forecasting and drought prediction, analyzing Transformer-, graph-, and attention-based architectures to identify suitable modeling strategies.
- Designed candidate deep learning architectures integrating spatial attention with temporal sequence modeling for nationwide drought forecasting.

PUBLICATIONS & PREPRINTS

Query Expansion and Key Specialization in Transformer Attention Geometry

Vidit Gupta, Siddhesh Nadkarni, Mihik Chaudhari, Vinaya Sawant, Prachi Satam. | *Under Review*

Residual-Stream Geometry in Transformer Language Model

Vidit Gupta, Pranay Khandagle, Neha Katre, Sweedle Machado. | *Under Review*

SELECTED PROJECTS

Attention Geometry & Training Dynamics in Transformers | *PyTorch, Mech Interp*

Mar 2026 – Jun 2026

- Instrumented a 4M-parameter GPT-style transformer with 18 geometric metrics tracking attention entropy, effective rank, covariance spectra, Q/K/V statistics, and gradient dynamics across 87 epochs.
- Observed divergent query/key geometry despite shared projections, spectral redistribution in W_q , and increasing pre-softmax rank while post-softmax attention remained low-rank due to softmax concentration.
- Developed a reproducible measurement stack generating per-epoch JSONL logs and automated visualization for transformer geometry analysis.

Language Model Stack: From Training to Agents | *PyTorch, LLM Systems*

Jul 2026 – Present

- Trained a 50M-parameter decoder-only transformer from scratch with custom BPE tokenization, RoPE, RMSNorm, grouped-query attention, SwiGLU Mixture-of-Experts, weight tying, and cosine learning-rate decay.
- Built a token-level inference runtime for Qwen3 models implementing KV cache, greedy, beam, top- k , nucleus, min- p , and locally typical sampling without relying on `model.generate()`.
- Extended the inference runtime into a tool-using agent capable of interrupting decoding, invoking external tools, preserving KV cache across tool calls, and resuming generation.

Representation Geometry: Hidden State Analysis | *PyTorch, Hugging Face*

Jun 2026 – Present

- Developed and published a model-agnostic PyTorch library for transformer representation analysis using activation hooks, streaming covariance estimation, and reproducible artifact generation.
- Implemented spectral, statistical, and subspace metrics including participation ratio, stable rank, entropy rank, linear CKA, and activation-space SVD analysis.
- Supported Hugging Face and native PyTorch models through automatic transformer-block discovery with modular analysis and bounded-memory statistical backends.

Torch-IsoFLOPs: Chinchilla-Style Scaling Law Framework | *PyTorch, Scaling Laws*

Jun 2026 – Jun 2026

- Developed a reusable framework for IsoFLOP scaling-law experiments under fixed compute budgets using Chinchilla-style methodology.
- Implemented automatic FLOP estimation, experiment scheduling, Huber regression, RANSAC-based curve fitting, and resumable experiment logging.
- Enabled analytical estimation of optimal parameter-data tradeoffs across compute budgets with automated visualization.

EXTRACURRICULAR ACTIVITIES

DJ InIT.ai

Dec 2025 – Present

Vice Chairperson

Mumbai, India

- Lead a student research club focused on machine learning, organizing technical sessions, paper reading groups, and research discussions.
- Coordinate research collaborations and mentor members in exploring and implementing machine learning research.

DJS Codestars

Sep 2024 – Jul 2026

Technical Member

Mumbai, India

- Authored competitive programming problems for weekly contests engaging 150+ participants
- Contributed to organizing CodeUnicode, collaborating with teams on contest preparation and website development.

GDG DJSCE

Oct 2025 – Jul 2026

ML Research Member

Mumbai, India

- First author of a research paper under review on redundancy reduction in transformer hidden-state representations.
- Participated in research paper reading sessions and technical discussions on modern machine learning.